- $\int dG(x) = G(x) + C$

令 $G(x) = F(\varphi(x))$，则

$dG(x) = F'(\varphi(x)) \cdot \varphi'(x)\,dx$，从而

$$\int F'(\varphi(x)) \cdot \varphi'(x)\,dx = F(\varphi(x)) + C$$

令 $F'(x) = f(x)$，则有

$$\int f(\varphi(x))\,\varphi'(x)\,dx = F(\varphi(x)) + C$$

若令 $u = \varphi(x)$，上式也可写成

$$\int f(u)\,du = F(u) + C$$

第一换元积分法

- 例1: $\int \frac{x}{1+x^2}\,dx$

$$= \frac{1}{2}\int \frac{2x\,dx}{1+x^2} = \frac{1}{2}\int \frac{d(x^2)}{1+x^2}$$

$$\overset{u=x^2}{=\!=} \frac{1}{2}\int \frac{du}{1+u} = \frac{1}{2}\ln(1+u)+C$$

$$= \frac{1}{2}\ln(1+x^2)+C$$

- 例2: $\int \tan x\,dx$

$$= \int \frac{\sin x}{\cos x}\,dx = -\int \frac{d(\cos x)}{\cos x}$$

$$= -\ln|\cos x|+C$$

- 例: $\int x^2 \sqrt[3]{1-x^3}\, dx$

$= \frac{1}{3} \int \sqrt[3]{1-x^3}\, d(x^3)$

$= -\frac{1}{3} \int \sqrt[3]{1-x^3}\, d(1-x^3)$

$\overset{u=1-x^3}{=\!=} -\frac{1}{3} \int u^{\frac{1}{3}} du = -\frac{1}{4} u^{\frac{4}{3}} + C$

$= -\frac{1}{4}(1-x^3)^{\frac{4}{3}} + C$

- 例: $\int \frac{dx}{x(3+2\ln x)}$

$= \int \frac{\frac{1}{x} dx}{3+2\ln x} = \int \frac{d(\ln x)}{3+2\ln x}$

$\overset{u=\ln x}{=\!=} \int \frac{du}{3+2u} = \frac{1}{2} \ln|3+2u| + C$

$= \frac{1}{2} \ln|3+2\ln x| + C$

- 例1: $\int \sin^2 x \cos x \, dx$

$= \int \sin^2 x \; d(\sin x) = \frac{1}{3}\sin^3 x + C$

- 例1 $\int \sin^3 x \, dx$

$= -\int \sin^2 x \, d(\cos x) = -\int (1-\cos^2 x) \, d(\cos x)$

$\xrightarrow{u=\cos x} -\int (1-u^2) \, du = -u + \frac{u^3}{3} + C$

$= -\cos x + \frac{\cos^3 x}{3} + C$

- 例1: $\int \sin 4x \cdot \cos 5x \, dx$

由于 $\sin\alpha \cdot \cos\beta = \frac{1}{2}\left(\sin(\alpha+\beta) + \sin(\alpha-\beta)\right)$

$\sin 4x \cdot \cos 5x = \frac{1}{2}(\sin 9x - \sin x)$

原式 $= \int \frac{1}{2}(\sin 9x - \sin x) \, dx$

$= -\frac{1}{18}\cos 9x + \frac{1}{2}\cos x + C$

- 例: $\int \tan^4 x \, dx$

$= \int \tan^2 x \cdot (\sec^2 x - 1) \, dx$

$= \int \tan^2 x \cdot \sec^2 x \, dx - \int \tan^2 x \, dx$

$= \int \tan^2 x \, d(\tan x) - \int (\sec^2 x - 1) \, dx$

$= \frac{1}{3}\tan^3 x - \tan x + x + C$

- 例: $\int \frac{dx}{a^2 + x^2} = \int \frac{1}{a^2} \cdot \frac{dx}{1 + (\frac{x}{a})^2}$

$= \frac{1}{a} \cdot \int \frac{d(\frac{x}{a})}{1 + (\frac{x}{a})^2} = \frac{1}{a} \arctan \frac{x}{a}$

书上例2.7. (2). (3) 自己看书.

例2.8 自己看书. (推荐解法2).

第二换元积分法：

若 $x=\varphi(t)$，$\varphi'(t)>0$ (<0)，则

$$\int f(x)\,dx=\int f(\varphi(t))\,d\varphi(t)$$

$$=\int f(\varphi(t))\cdot\varphi'(t)\,dt$$

换元：$x=\varphi(t)$.

- 这里 $\varphi'(x)>0$ (<0) 只是为了保证 x 与 t 一一对应，或者说 $\varphi(t)$ 可逆.
- 求出 $\int f(\varphi(t))\cdot\varphi(t)\,dt$ 之后应将 $t=\varphi^{-1}(x)$ 代回去，变成 x 的函数

三角代换：

例：$\int \sqrt{a^2-x^2}\,dx \quad (a>0)$.

解：这里 $-a \le x \le a$，从而可设

$x = a\sin t \quad . \quad t \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

原式 $= \int \sqrt{a^2 - a^2\sin^2 t}\, d(a\sin t)$

$= \int a\cos t \cdot a\cos t\, dt$

$= a^2 \int \frac{1+\cos 2t}{2} dt$

$= a^2 \cdot \left(\frac{t}{2} + \frac{1}{4}\sin 2t\right) + C$

$= \frac{a^2}{2} \cdot t + \frac{a^2}{2} \cdot \sin t \cdot \cos t + C$

$= \frac{a^2}{2} \cdot \arcsin\frac{x}{a} + \frac{1}{2} x \cdot \sqrt{a^2-x^2} + C$

- 例2.9 (2).(3) 可利用第95页公式(20)(21)来做.
- 例: $\int \frac{dx}{\sqrt{x(1-x)}}$

由于 $x \in (0,1)$. 不妨设

$x = \sin^2 t \quad t \in (0, \frac{\pi}{2})$

则 $\sin t = \sqrt{x}$, $t = \arcsin\sqrt{x}$

$$\text{原式} = \int \frac{d(\sin^2 t)}{\sin t \cdot \cos t} = \int 2\,dt$$

$$= 2t + C = 2\arcsin\sqrt{x} + C$$

这题也可以作代换 $t = \sqrt{x}$.

- 例2.10 (2) 自己看书.

- 无理代换

(1) $\int \frac{1}{1+\sqrt{x}}\,dx$

令 $t=\sqrt{x}$，则 $x=t^2$ $(t\geq 0)$

原式 $=\int \frac{1}{1+t}\,d(t^2)=2\int \frac{t\,dt}{1+t}$

$=2\int\left(1-\frac{1}{1+t}\right)dt=2t-\ln(1+t)+C$

$=2\sqrt{x}-\ln(1+\sqrt{x})+C$

- 例: $\int \frac{dx}{\sqrt{x}-\sqrt[3]{x^2}}$

解: 令 $\sqrt[6]{x}=t$，则 $x=t^6$ $(t>0)$

$$\int \frac{d(t^6)}{t^3-t^4}=6\int \frac{t^2\,dt}{1-t}$$

$=6\int\left(-t-1+\frac{1}{1-t}\right)dt$ (略).

- 例：$\int \frac{x}{\sqrt{1+\sqrt[3]{x^2}}}dx$

解：令 $t=\sqrt{1+\sqrt[3]{x^2}}$，则 $(t\geq 1)$

$$t^2=1+\sqrt[3]{x^2},\quad t^2-1=\sqrt[3]{x^2}$$

$$(t^2-1)^3=x^2$$

$$\text{原式}=\frac{1}{2}\int\cdot\frac{dx^2}{\sqrt{1+\sqrt[3]{x^2}}}=\frac{1}{2}\int\frac{d(t^2-1)^3}{t}$$

$$=3\int(t^2-1)^2dt=\cdots\quad(\text{略})$$

例 2.11 (3) 自己看书.

倒代换

- 例：$\int \frac{\sqrt{a^2-x^2}}{x^4}\,dx$

$$= \int \frac{\sqrt{a^2-x^2}}{x^2}\cdot\frac{dx}{x^2}$$

$$= -\int \frac{\sqrt{a^2-x^2}}{x^2}\cdot d\left(\frac{1}{x}\right)$$

$$\overset{t=\frac{1}{x}}{=\!=\!=} -\int t^2\cdot\sqrt{a^2-\frac{1}{t^2}}\,dt \quad \left(\begin{matrix}\text{不妨}\ t>0\\ x>0\end{matrix}\right)$$

$$= -\int t\cdot\sqrt{a^2t^2-1}\,dt$$

$$= -\frac{1}{2a^2}\int \sqrt{a^2t^2-1}\,d\left(a^2t^2-1\right)$$

$$= -\frac{1}{3a^2}\left(a^2t^2-1\right)^{\frac{3}{2}}+C = -\frac{\left(a^2-x^2\right)^{\frac{3}{2}}}{3a^2x^3}+C$$

当 $x<0$ 时，类似可得相同的结果. （$x>0$）

- 例：$\int \frac{dx}{x\cdot\sqrt{x^4+x^2}}$

$$=\frac{1}{2}\int \frac{d(x^2)}{x^2\sqrt{x^4+x^2}} \xrightarrow{u=x^2} \frac{1}{2}\int \frac{du}{u\cdot\sqrt{u^2+u}}$$

$$=\frac{1}{2}\int \frac{du}{u^{\frac{3}{2}}\cdot(1+u)^{\frac{1}{2}}} = -\int \frac{du^{-\frac{1}{2}}}{(1+u)^{\frac{1}{2}}}$$

$$\xrightarrow{v=u^{-\frac{1}{2}}} -\int \frac{dv}{\sqrt{1+\frac{1}{v^2}}} = -\int \frac{v\,dv}{\sqrt{1+v^2}}$$

$$=-\frac{1}{2}\int \frac{d(v^2+1)}{\sqrt{1+v^2}} = -\sqrt{1+v^2}+C$$

$$=-\sqrt{1+\left(u^{-\frac{1}{2}}\right)^2}+C=-\sqrt{1+\frac{1}{u}}+C$$

$$=-\sqrt{1+\frac{1}{x^2}}+C$$

分部积分法：

$$d(u\cdot v)= u\cdot dv+v\cdot du$$

从而

$$\int d(u\cdot v)=\int u\,dv+\int v\,du$$

$$\int u\,dv=\int d\cdot(u\cdot v)-\int v\,du$$

即

$$\int u\,dv= u\cdot v-\int v\,du$$

分部积分公式.

- 例：$\int x \sin x \, dx$

$= -\int x \, d(\cos x)$

而 $\int x \, d(\cos x) = x \cdot \cos x - \int \cos x \, dx$

$= x \cdot \cos x - \sin x + C$

故 $\int x \sin x \, dx = -x \cos x + \sin x + C$

- 例：$\int (x^2 + 3x + 1) \ln x \, dx$

$= \int \ln x \, d\left(\frac{x^3}{3} + \frac{3}{2}x^2 + x\right)$

$= \left(\frac{x^3}{3} + \frac{3}{2}x^2 + x\right) \cdot \ln x - \int \left(\frac{x^3}{3} + \frac{3}{2}x^2 + x\right) d(\ln x)$

$= \left(\frac{x^3}{3} + \frac{3}{2}x^2 + x\right) \cdot \ln x - \int \left(\frac{x^2}{3} + \frac{3}{2}x + 1\right) dx$

$= \left(\frac{x^3}{3} + \frac{3}{2}x^2 + x\right) \cdot \ln x - \left(\frac{x^3}{9} + \frac{3}{4}x^2 + x\right) + C$

例：$\int x^2 e^x dx$

$= \int x^2 d(e^x) = x^2 \cdot e^x - \int e^x d(x^2)$

$= x^2 \cdot e^x - 2\int e^x \cdot x\, dx$

$= x^2 \cdot e^x - 2\int x \cdot d(e^x)$

$= x^2 \cdot e^x - 2 \cdot x \cdot e^x + 2\int e^x dx$

$= x^2 e^x - 2xe^x + 2e^x + C$

$= (x^2 - 2x + 2)e^x + C$

- 对于 $\int x^n \cdot e^x \cdot dx$ 都可以类似地处理. 令 $I_n = \int x^n e^x dx$, 可以建立递推公式. (自己练习)

- 例 2.13 (2) 自己看书.

一类特殊的积分

- 例 $\int e^x \cdot \sin x \, dx$

$$= \int \sin x \cdot d(e^x) = e^x \cdot \sin x - \int e^x d(\sin x)$$

$$= e^x \cdot \sin x - \int e^x \cos x \cdot dx$$

$$= e^x \cdot \sin x - \int \cos x \cdot d(e^x)$$

$$= e^x \cdot \sin x - \cos x \cdot e^x + \int e^x \cdot d(\cos x)$$

$$= e^x \sin x - e^x \cdot \cos x - \int e^x \cdot \sin x \cdot dx$$

故 $\int e^x \cdot \sin x \cdot dx = \frac{1}{2}\left(e^x \cdot (\sin x - \cos x)\right)$

- 例2.14 (2). (3) 方法类似. 见书.

特别地, 在(3)中, 令 $x = a\tan t$, $t \in (-\frac{\pi}{2}, \frac{\pi}{2})$. 则 $\sqrt{x^2 + a^2} = a\sec t$.

$$\int \sqrt{x^2+a^2}\,dx = \int a\sec t\, d(a\cdot\tan t)$$

$$= a^2\int \sec^3 t\, dt,\ (3)\text{转化成了}(2).$$

- 例：建立不定积分的递推公式.

$$I_n = \int \frac{1}{(a^2+x^2)^n}dx \quad (a>0),\ n\in\mathbb{Z}^+.$$

解：$I_n = \int \frac{1}{(a^2+x^2)^n}dx$

$$= x\cdot\frac{1}{(a^2+x^2)^n} - \int x\cdot d\frac{1}{(a^2+x^2)^n}$$

$$= x\cdot\frac{1}{(a^2+x^2)^n} + 2n\cdot\int\frac{\cdot x^2\,dx}{(a^2+x^2)^{n+1}}$$

$$= x\cdot\frac{1}{(a^2+x^2)^n} + 2n\cdot\left(\int\frac{dx}{(a^2+x^2)^n} - \int\frac{a^2}{(a^2+x^2)^{n+1}}dx\right)$$

$$= \frac{x}{(a^2+x^2)^n} + 2n I_n - 2n \cdot a^2 \cdot I_{n+1}$$

故 $I_{n+1} = \frac{1}{2n \cdot a^2} \frac{x}{(a^2+x^2)^n} + \frac{2n-1}{2n \cdot a^2} \cdot I_n$

当 $n=1$ 时，

$$I_1 = \int \frac{1}{a^2+x^2} dx = \frac{1}{a} \arctan \frac{x}{a}$$

- 例：设 $f(x)$ 的原函数为 $\frac{\sin x}{x}$，

求 $\int x f'(x) dx$.

解：$f(x) = \left(\frac{\sin x}{x}\right)' = \frac{\cos x \cdot x - \sin x}{x^2}$

$$\int x \cdot f'(x) dx = \int x \, d f(x)$$

$$= x \cdot f(x) - \int f(x) dx = x \cdot f(x) - \frac{\sin x}{x} + C$$

$$= \frac{\cos x \cdot x - \sin x}{x} - \frac{\sin x}{x} + C = \cos x - \frac{2\sin x}{x} + C$$

作业：

习题4-2A类

2. (2). (6). (9). (10).

3. (1). (3). (6). (11).

4. (2). (3).

习题4-2B类

1. (12). (17)

2. (1).